Sheet 02: Principles of Sectioning and Cutting Planes

TOPIC: SECTIONAL ELEVATIONS, HATCHING CONVENTIONS & THREAD PROFILES

1. CUTTING PLANE PRINCIPLES:

- A cutting plane is represented by a thin chain line with thick ends and arrows indicating viewing direction.
- Material cut by the plane is hatched using thin continuous lines at 45 degrees to the main outlines.
- Distance between hatching lines should be uniform (1.5mm to 3mm).

2. TYPES OF SECTIONS:

- Full Section: Cutting plane passes entirely through the component.
- Half Section: Applied to symmetrical objects (half exterior, half interior).
- Offset Section: Cutting plane bends at right angles to pass through multiple features.
- Revolved & Removed Sections: For spokes, ribs, shafts, and crane hooks.

3. STANDARD FASTENERS:

- Hexagonal Bolt & Nut: D = nominal diameter, Head thickness = 0.8D, Across flats = 1.5D + 3mm.
- Metric Screw Thread (ISO 68-1): 60 degree included angle, P = pitch.

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